

# Raphael Baer-Way

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## Research Interests

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Core-collapse supernovae, Stellar mass loss, Massive stars, Circumstellar interaction, Multiwavelength observations

## Education

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**Ph.D.**, Astronomy, University of Virginia 2023–2028 (expected)

*Advisors: Prof. Maryam Modjaz, Dr. Poonam Chandra*

*GPA: 4.00/4*

**B.A.**, Astronomy and Physics, University of California, Berkeley 2018–2022

*Research Advisor: Prof. Alex Filippenko*

*GPA: 3.89/4 (151 credits)*

## Honors & Awards

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NSF Graduate Research Fellowship 2025–2029

Jefferson Teaching Award, University of Virginia Astronomy 2025

Virginia Space Grant Consortium Graduate Fellowship 2024–2025

American Astronomical Society Travel Grant 2024

Outstanding Graduate Student Instructor Award, UC Berkeley 2022

Shared Chambliss Astronomy Achievement Student Award, AAS 2021

Dean's List, UC Berkeley 2018

Award for Innovation and Creativity in Science, Rochester Institute of Technology 2017

## Refereed Publications

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**16 total; 3 first-author, 2 second-author (h-index=10, total citations=198)**

**Baer-Way et al.**, “The First Radio View of a Type Ibn Supernova in SN 2023fyq: Understanding the Mass-loss History in the Last Decade before the Explosion” 2025

*The Astrophysical Journal*, 995:L49, December 2025

**Baer-Way et al.**, “A Multiwavelength Autopsy of the Interacting IIn Supernova 2020ywx: Tracing its Progenitor Mass-Loss History for 100 Years before Death” 2025

*The Astrophysical Journal*, 983:101, April 2025

**Baer-Way et al.**, “A Snapshot Survey of Nearby Supernovae with the Hubble Space Telescope” 2024

*The Astrophysical Journal*, 964:2, April 2024

**Kumar, Baer-Way et al.**, “A Multiwavelength View of the Nearby Calcium-strong Transient SN 2025coe in the X-Ray, Near-infrared, and Radio Wavebands” 2026

*The Astrophysical Journal*, 1005:98, July 2026

**Hillenkamp, Baer-Way et al.**, “Fading Echoes of Interaction: Probing Centuries of Preexplosion Mass Loss in Four Type IIn Supernovae” 2026

*The Astrophysical Journal*, 1001:111, April 2026

**Ravi, A., Kumar S., Baer-Way et al.**, “The Double-peaked Calcium-strong SN 2025coe: Progenitor Constraints from Early Interaction and Ejecta Asymmetries” 2026

*The Astrophysical Journal*, 1003:33, May 2026

Smith et al., incl. **Baer-Way, R.**, “JWST Spectra Indicate a Large Mass of Postshock Dust Formed by SN 2010jl” 2026  
*The Astrophysical Journal*, 1006:224, August 2026

Crawford et al., incl. **Baer-Way, R.**, “Peaky Finders: Characterizing Double-peaked Type IIb Supernovae in Large-scale Live-stream Photometric Surveys” 2025  
*The Astrophysical Journal*, 989:192, August 2025

Pellegrino et al., incl. **Baer-Way, R.**, “The X-Ray Luminous Type Ibn SN 2022ablq: Estimates of Preexplosion Mass Loss and Constraints on Precursor Emission” 2024  
*The Astrophysical Journal*, 977:2, December 2024

Pichay et al., incl. **Baer-Way, R.**, “SN 2021ukt: A Transitional Supernova with a Short Plateau and Persistent Interaction” 2026  
*The Astrophysical Journal*, 1007:13, August 2026

Yesmin et al., incl. **Baer-Way, R.**, “Spectral Dataset of Young Type Ib Supernovae and Their Time Evolution” 2025  
*A&A*, 693, February 2025

Van Dyk, deGraw, **Baer-Way et al.**, “The Disappearance of Six Supernova Progenitors” 2022  
*MNRAS*, 519, 471–482, December 2022

Zheng et al., incl. **Baer-Way, R.**, “The Lick Observatory Supernova Search Follow-up Program: Photometry Data Release of 70 SESNe” 2022  
*MNRAS*, 512, 3195–3214, March 2022

Aryan et al., incl. **Baer-Way, R.**, “Progenitor Mass Constraints for the Type Ib Intermediate-Luminosity SN 2015ap and the Highly Extinguished SN 2016bau” 2021  
*MNRAS*, 505, 2530–2547, May 2021

Alvarado et al., incl. **Baer-Way, R.**, “Searching for tidal orbital decay in hot Jupiters ” 2024  
*MNRAS* 534, 800–813, September 2024

Murakami et al., incl. **Baer-Way, R.**, “PIPS, an advanced platform for period detection in time series - I. Fourier-likelihood periodogram and application to RR Lyrae stars” 2022  
*MNRAS* 514, 4489–4505, June 2022

## Teaching Experience/Outreach

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Head TA, UVA Astronomy 2024–2025  
 Organized TA assignments and duties for 10 TAs for the academic year

TA, Astro 3130 (Advanced Optical Astronomy), UVA 2024–2025  
 Taught lab sessions and organized observing sessions for 60 students; received teaching award

Volunteer & Meeting Leader, Dark Skies Bright Kids UVA 2023–  
 Led bi-weekly sessions teaching astronomy to 20 3rd–5th graders; helped organize outreach for 100s of people at elementary schools and community centers

TA/UGSI, Astron 10 (Intro to General Astronomy), UC Berkeley 2022  
 Taught two sections of 40 students; received teaching award

TA, Astron 9 (Intro to Python for Astrophysicists), UC Berkeley 2022  
 Co-taught course of 30 students; graded and led lecture and supplemental discussions

Python Instructor (student-led DeCal), UC Berkeley 2020–2022  
 Instructor for Python DeCal course for astrophysicists for > 100 students, 4 semesters

## Mentorship

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Altony Foote, UVA undergraduate 2025  
 Work on late-time lightcurves of stripped-envelope Supernovae

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|--------------------------------------------------------------------------------------------------|-----------|
| Michaela Schwab, UVA graduate student                                                            | 2025-     |
| <i>Work on optical analysis of Ic-bl/transition supernovae</i>                                   |           |
| Elizabeth Hillenkamp, NRAO Postbaccalaureate Researcher                                          | 2024-2026 |
| <i>Work on strongly-interacting hydrogen-rich supernovae, Second-author paper published 2026</i> |           |
| Maanya Jarayam, NRAO Summer student                                                              | 2026      |
| <i>Work on radio lightcurves of type IIb Supernovae</i>                                          |           |

## Selected Press

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| UVA/NRAO Press Release on first radio detection of rare supernova | 2026 |
| <i>UVA Article, NRAO Press Release</i>                            |      |

## Conference Contributions

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### Talks

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|----------------------------------------------------------------------------------------------------------------|------|
| “Multiwavelength insights into helium-rich, hydrogen-poor supernovae” Rise Time, Lafayette, Indiana            | 2026 |
| “MMT insights into core-collapse supernovae” MMT Symposium, Tuscon, Arizona                                    | 2026 |
| “Multiwavelength views of two strongly interacting supernovae” Time Domain Soiree, New Brunswick, New Jersey   | 2025 |
| “Multiwavelength views of two strongly interacting supernovae” Supernovae Group Meeting, IIA, Bangalore, India | 2025 |

### Posters

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| “Multiwavelength views of helium-rich supernovae,” Future Landscape of Astrophysical Transients, Turku, Finland | 2026 |
| “A Multi-Wavelength Autopsy of the Interacting Supernova 2020ywx,” Supernova Remnants III, Chania, Greece       | 2024 |
| “A Snapshot Survey of Nearby Supernovae with the Hubble Space Telescope,” AAS Meeting 242, Seattle, WA          | 2023 |

## Telescope Proposals/Observing Experience

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### Very Large Array

*5 proposals as PI, ~ 40 total hours, > 10 proposals as Co-I, ~ 100 total hours*

### Giant Meterwave Radio Telescope

*3 proposals as PI, ~ 50 total hours, 6 proposals as Co-I, ~ 100 total hours*

### Chandra X-Ray Observatory

*1 Proposal as Co-I, 50 ks*

### James Webb Space Telescope

*1 Archival proposal as Co-I*

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| Nickel Telescope, Lick Observatory | 2019–2022 |
| <i>Observed 15+ times</i>          |           |

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| Clay/Baade Telescopes, Las Campanas Observatory                            | 2023- |
| <i>Observed 5+ nights with optical IMACS and infrared FIRE instruments</i> |       |

## References

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|---------------------------------------------|---------------------|
| Prof. Maryam Modjaz, University of Virginia | vru7qe@virginia.edu |
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**Dr. Poonam Chandra, NRAO**  
**Prof. Alex Filippenko, UC Berkeley**

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